

Network Calculus

Henrik Schiøler

Network Calculus

- A theory for the determination of backlogs and delays in networks of network elements.
- Network elements comprise processors, servers, switches,...
- Network elements are modeled as buffer and service elements



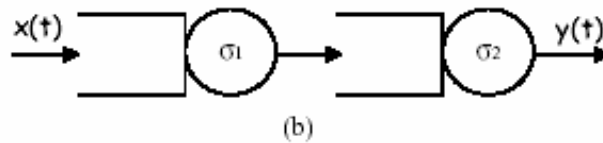
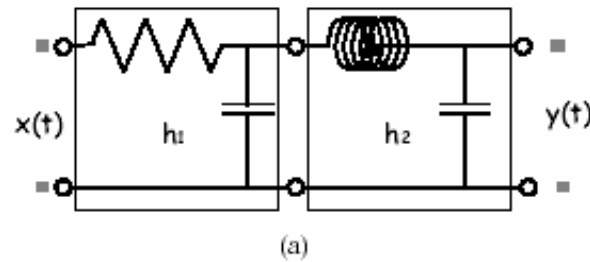
- Based on input- and server-characteristic backlog- and delay-bounds as well as output characteristics are computed
- All models are lossless (no data lost and none produced)

Background

- Queueing Theory gives probabilistic results
- Critical applications need hard bounds
- Queueing theory extends only partially to networks.
- Scheduling theory accounts only for CPU sharing and bounded blocking.
- Periodic studies do not account for bursty traffic.

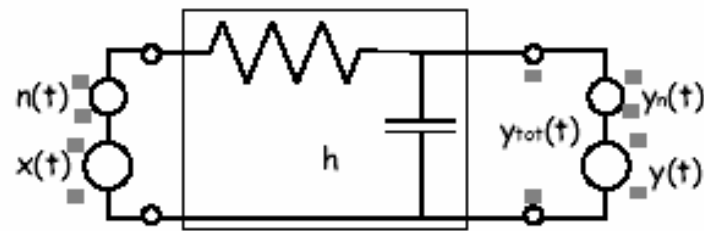
Concatenation of Network Elements

$$h(t) = (h_1 \otimes h_2)(t) = \int_0^t h_1(t-s)h_2(s)ds$$

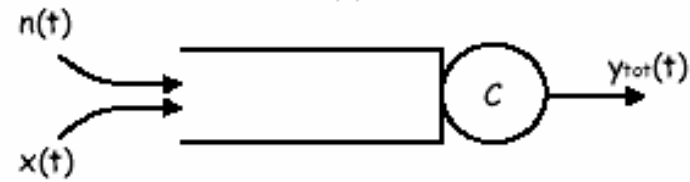


$$\sigma(t) = (\sigma_1 \otimes \sigma_2)(t) = \inf_{0 \leq s \leq t} [\sigma_1(t-s) + \sigma_2(s)]$$

Converging Flows

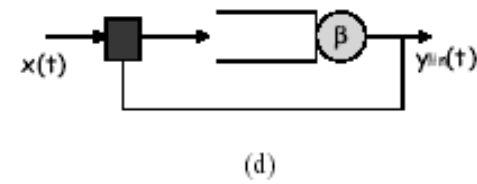
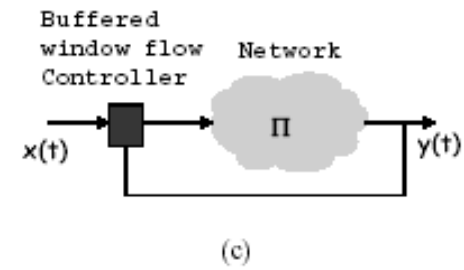
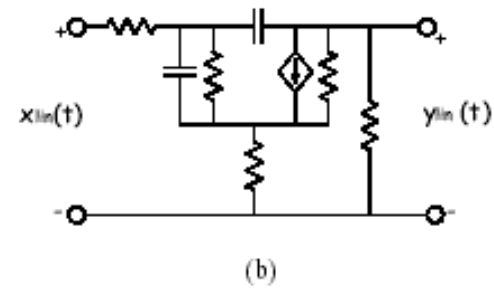
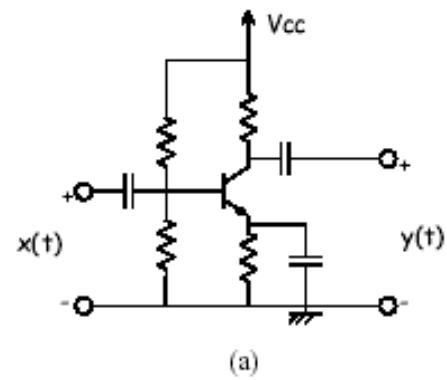


(a)



(b)

Feedback

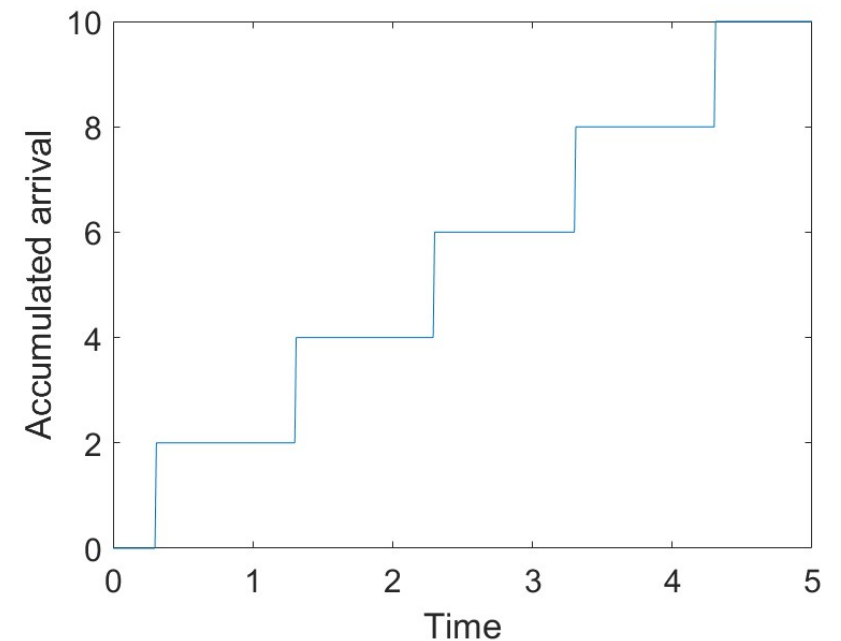


Accumulated flow

- $R(t)$: amount of data/work received in $[0,t]$
- $R(0)=0$
- R is non-decreasing: $R(t) \geq R(s)$ for $t \geq s$
- R may be discontinuous
- For periodic arrivals:

$$R(t) = k \left\lceil \frac{t - O}{T} \right\rceil$$

- $0 \leq O < T$



Arrival Curve

DEFINITION 1.2.1 (Arrival Curve). *Given a wide-sense increasing function α defined for $t \geq 0$ we say that a flow R is constrained by α*

if and only if for all $s \leq t$:

$$R(t) - R(s) \leq \alpha(t - s)$$

We say that R has α as an arrival curve, or also that

R is α -smooth.

Bounding every future

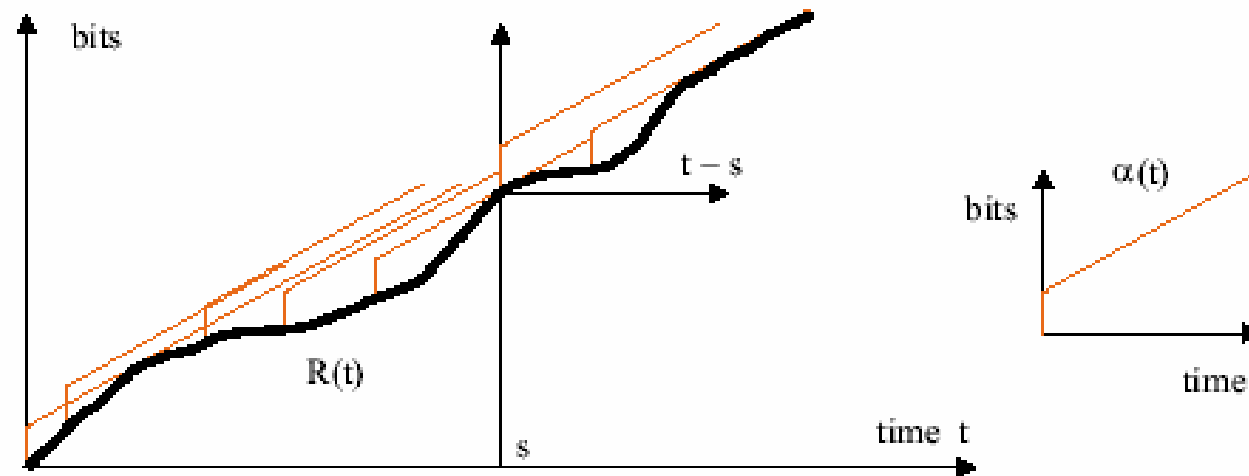


Figure 1.3: Example of Constraint by arrival curve, showing a cumulative function $R(t)$ constrained by the arrival curve $\alpha(t)$.

Spacing as a constraint

- $v_{T,\tau} = \left\lceil \frac{t+\tau}{T} \right\rceil$
- $v_{T,\tau}(t) = v_{T,0}(t + \tau)$

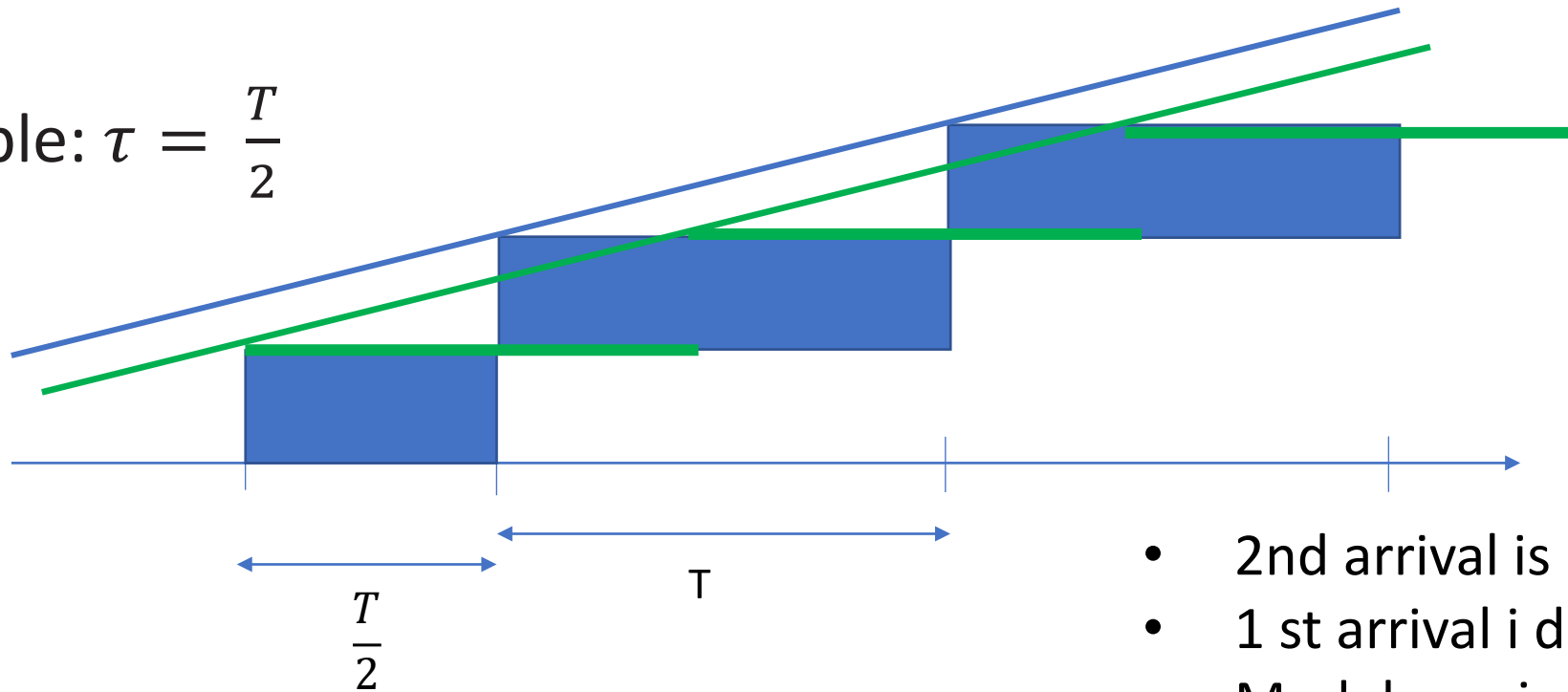
PROPOSITION 1.2.1

- *Consider a flow, with cumulative function $R(t)$, that*
- *generates packets of constant size equal to k data units,*
- *with instantaneous packet arrivals.*
- *Call t_n the arrival time for the n th packet.*
- *The following two properties are equivalent:*

1. *for all m, n , $t_{m+n} - t_m \geq nT - \tau$*
2. *the flow has $k v_{T,\tau}$ as an arrival curve*

Spacing as a constraint

- for all m, n , $t_{m+n} - t_m \geq nT - \tau$
- $0 < \tau$
- Example: $\tau = \frac{T}{2}$

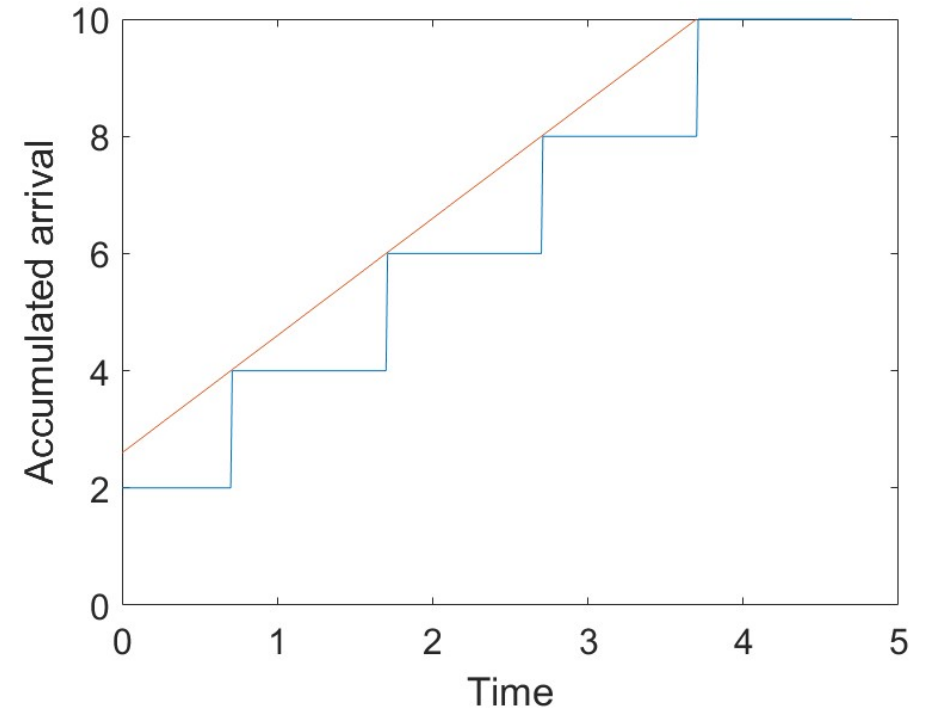


- 2nd arrival is not early
- 1st arrival is delayed
- Models periodic source with varying transmission delay

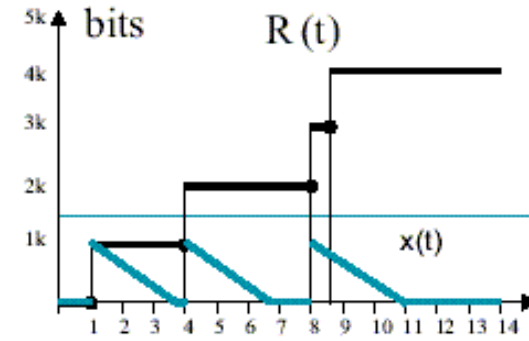
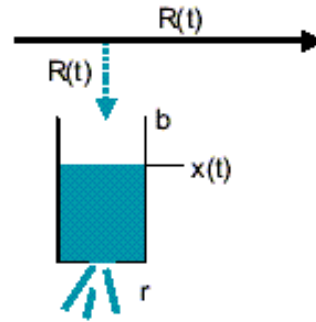
An equivalence

- $\gamma_{r,b}(t) = r t + b$
- PROPOSITION 1.2.2.
- $R(t)$ that generates packets of constant size equal to k data units,
- with instantaneous packet arrivals.
- Let $r = k/T$ and $b = k(\tau/T + 1)$.

It is equivalent to say that R is constrained by $\gamma_{r,b}$ or by $k \nu_T$,



The Leaky Bucket



DEFINITION 1.2.2

A Leaky Bucket Controller is a device that analyzes the data on a flow $R(t)$ as follows:

- *There is a pool (bucket) of fluid of size b .*
- *The bucket is initially empty.*
- *The bucket has a hole and leaks at a rate of r units of fluid per second when it is not empty.*
- *Data from the flow $R(t)$ has to pour into the bucket an amount of fluid equal to the amount of data.*
- *Data that would cause the bucket to overflow is declared non-conformant, otherwise the data is declared conformant*

Leaky buckets - properties

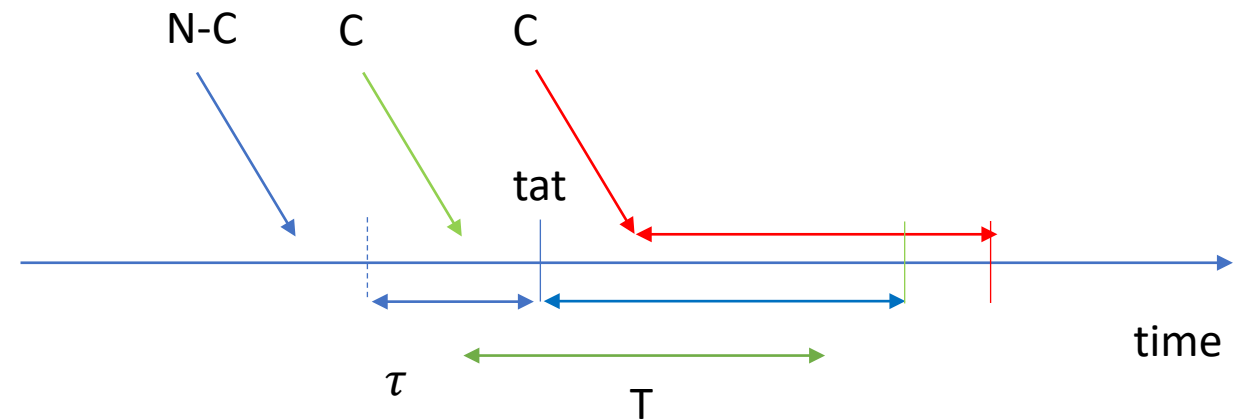
PROPOSITION 1.2.3. *A leaky bucket controller with leak rate r and bucket size b forces a flow to be constrained by the arrival curve $\gamma_{r,b}$, namely:*

- 1. the flow of conformant data has $\gamma_{r,b}$ as an arrival curve;*
- 2. if the input already has $\gamma_{r,b}$ as an arrival curve, then all data is conformant.*

The Generic Cell Rate Algorithm

DEFINITION 1.2.3 (GCRA (T, τ)). *The Generic Cell Rate Algorithm (GCRA) with parameters (T, τ) is used with fixed size packets, called cells, and defines conformant cells as follows. It takes as input a cell arrival time t and returns `result`. It has an internal (static) variable `tat` (theoretical arrival time).*

```
initially, tat = 0
when a cell arrives at time t, then
if (t < tat - tau)
    result = NON-CONFORMANT;
else {
    tat = max(t, tat) + T;
    result = CONFORMANT;
}
```



GCRA equivalence

PROPOSITION 1.2.4.

Consider a flow:

- with cumulative function $R(t)$
- that generates packets of constant size equal to k data units
- with instantaneous packet arrivals.
- The following two properties are equivalent:
 1. the flow is conformant to $\text{GCRA}(T, \tau)$
 2. the flow has $k\nu_{T,\tau}$ as an arrival curve

GCRA and Leaky Bucket

Corollary 1.2.1. For a flow with packets of constant size, satisfying the $GCRA(T, \tau)$ is equivalent to satisfying a leaky bucket controller, with rate r and burst tolerance b given by:

$$b = (\tau/T + 1)k$$

$$r = k/T$$

Summary of Equivalences

- Staircase arrival constraint: $k \nu_{T,\tau}$
- Affine arrival constraint: $\gamma_{r,b}$
- Leaky Bucket: (rate: r , bucket size: b)
- Generic Cell Rate Algorithm: $\text{GCRA}(T, \tau)$

(all under the assumption of constant packet size " k " and instantaneous arrival.)

A first result

LEMMA 1.2.2.

- *Consider a buffer served at a constant rate r .*
- *Assume that the buffer is empty at time 0.*
- *The input is described by the cumulative function $R(t)$.*
- *If there is no overflow during $[0, t]$, the buffer content x at time t is given by:*

$$x(t) = \sup_{s \leq t} \{R(t) - R(s) - r(t - s)\}$$

Proof

$$x(t) = \sup_{s \leq t} \{R(t) - R(s) - r(t - s)\}$$

First:

$$x(t) = x(s) + I(s, t) - O(s, t)$$

$$I(s, t) = R(t) - R(s)$$

$$O(s, t) \leq r(t - s) \quad (\text{because of possible idle periods})$$

$$x(s) \geq 0$$

$$x(t) \geq R(t) - R(s) - r(t - s)$$

Proof

$$x(t) = \sup_{s \leq t} \{R(t) - R(s) - r(t - s)\}$$

Next:

Assume $x(t)=0$

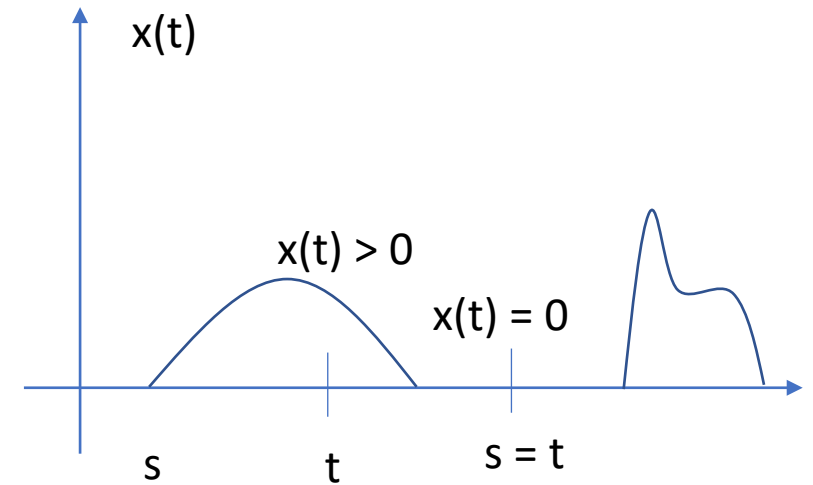
Set $s=t$, then

$$x(t) = R(t) - R(s) - r(t - s) = 0$$

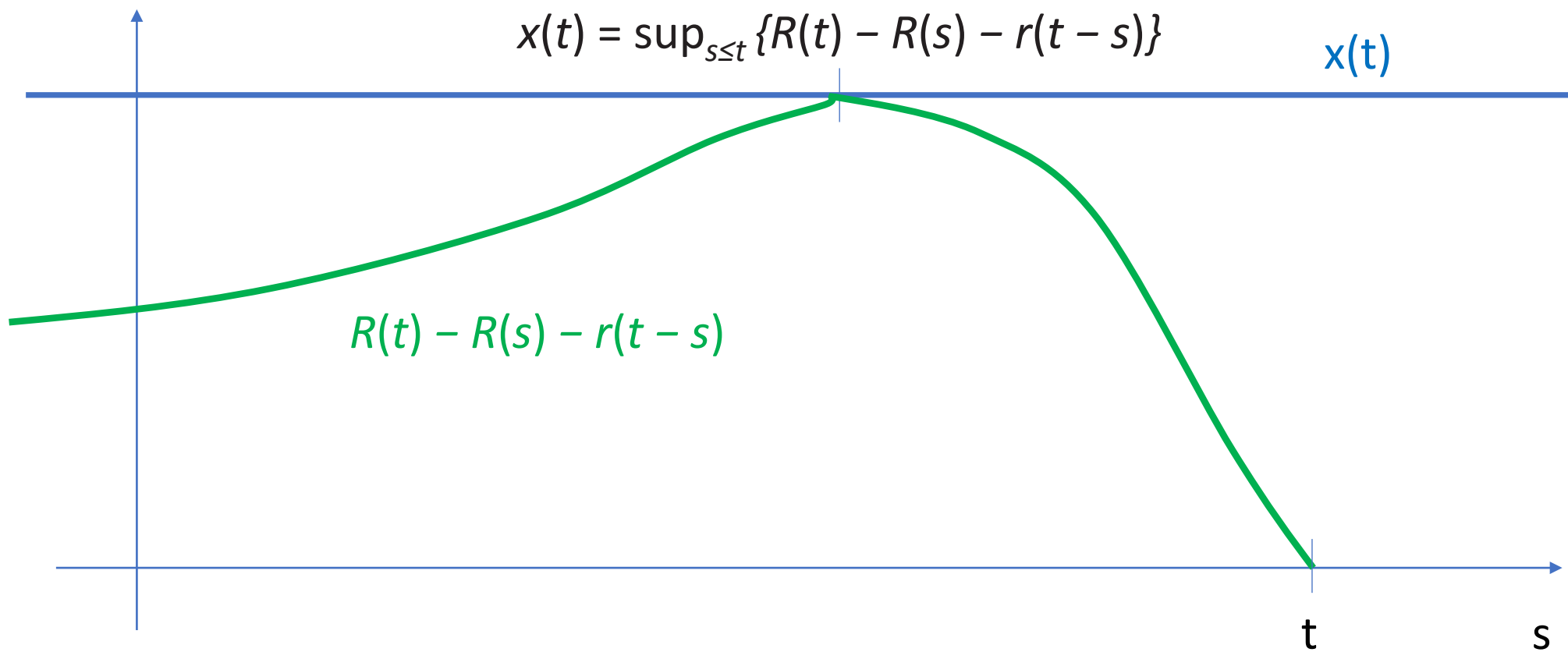
Assume $x(t) > 0$

Set $s < t$, latest start of busy period

$$x(t) = R(t) - R(s) - r(t - s)$$



Proof



Immediate Consequence

- Let

$$R(t) - R(s) \leq \alpha(t - s)$$

- Then

$$\begin{aligned} x(t) &= \sup_{s \leq t} \{R(t) - R(s) - r(t - s)\} \\ &\leq \sup_{s \leq t} \{\alpha(t - s) - r(t - s)\} \\ &= \sup_{0 \leq t-s} \{\alpha(t - s) - r(t - s)\} \\ &= \sup_{0 \leq d} \{\alpha(d) - r(d)\} \end{aligned}$$

Finding maximum backlog

